

Survival Analysis Methods Correcting for Treatment Switching Effects in RCTs: Theory and SAS/R Code (Part II)

ASA short course recap

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Recap

Why Treatment-Switching Is a Problem

- Treatment switching is common in oncology trials
e.g., progression → crossover to active therapy
- Leads to non-compliance and post-randomization confounding
- ITT preserves randomization but **treatment effect is diluted or biased**
 - One-way Switching
 - Two-way Switching
- Need causal methods to reconstruct a “no-switching” counterfactual

Four Causal Inference Methods

Marginal Structural Model (MSM)

Inverse Probability of Censoring Weighting (IPCW)

Rank Preserving Structure Failure Time (RPFST) Model

Two Stage Estimation (TSE)

Recap

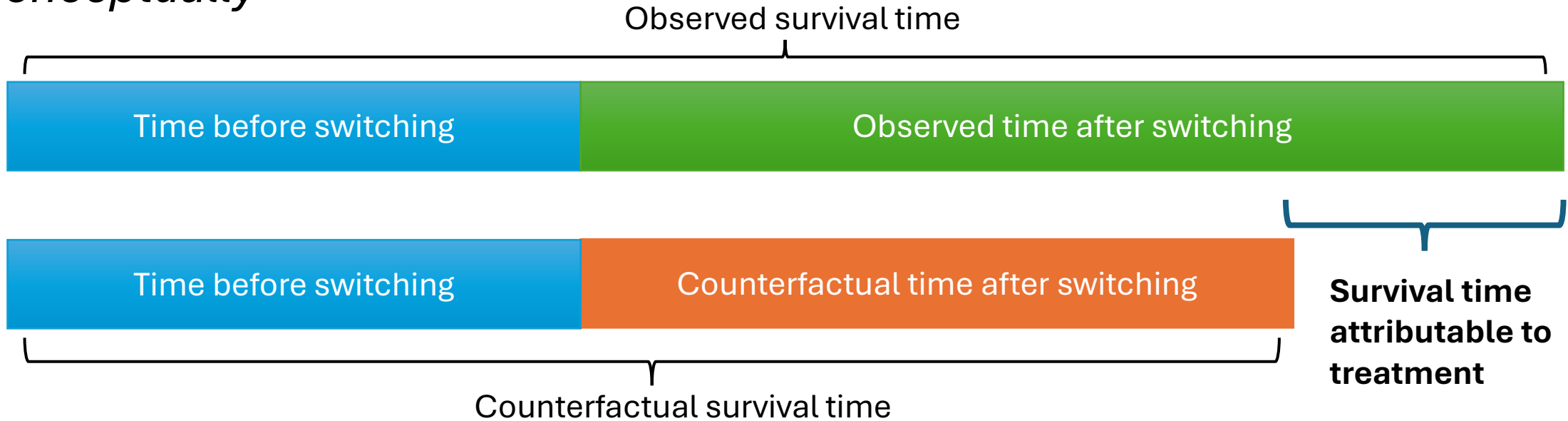
Weighting Methods

- MSM handles time-dependent confounding
- IPCW addresses informative censoring
- Both are **weighting-based methods** (inverse-probability weighting)
— create a pseudo-population where treatment & censoring behave “as-if random”
- Assumptions:

Method	Key Assumptions
MSM	1) No unmeasured time-varying confounding 2) Treatment Positivity 3) Correct treatment model 4) Consistency
IPCW	1) No unmeasured affecting censoring 2) Censoring positivity 3) Correct censoring model 4) Consistency

Rank Preserving Structure Failure Time Model (RPSFTM)

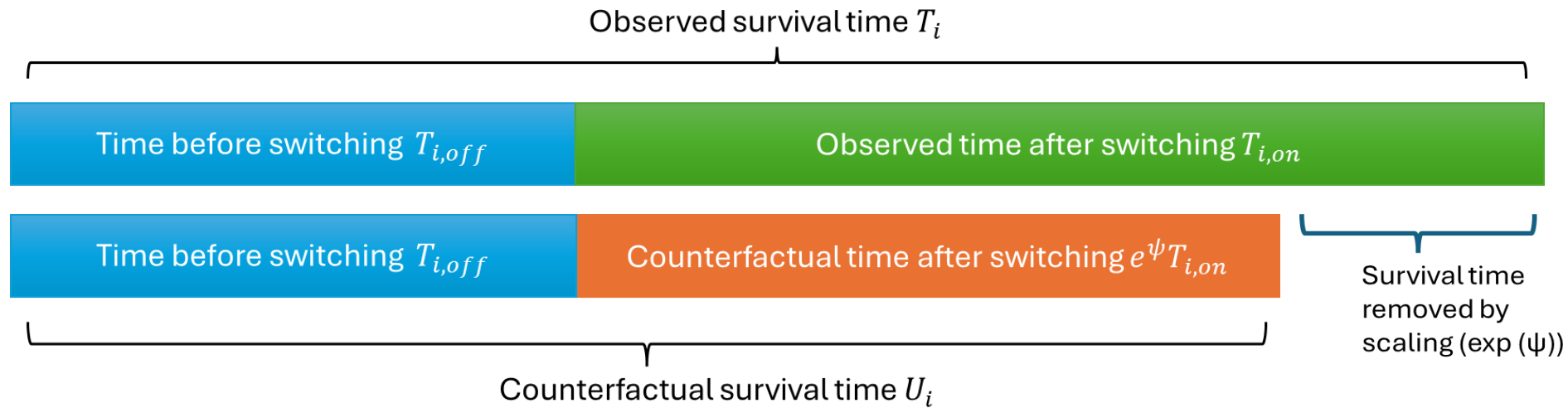
Conceptually



- **Observed survival = Time before switching + Time after switching**
 - Treatment can only affect survival after it begins
- **For the time after switching, RPSFTM asks a “what if” question:**
 - *What if this patient had never switched to the alternative treatment*
 - **If treatment is beneficial, survival during the treated period is longer than it would have been**
 - RPSFTM removes this benefit by shortening the post-treatment time
- **RPSFTM then finds the amount of shortening that makes the two randomized groups align**

Rank Preserving Structure Failure Time Model (RPSFTM)

Mathematical Formulation



- **Observed Survival Time:** $T_i = T_{i,off} + T_{i,on}$
Where $T_{i,off}$: time **before receiving active treatment**; $T_{i,on}$: time **on active treatment**
- **Counterfactual Survival Time:** $U_i(\psi) = T_{i,off} + T_{i,on} \exp(\psi)$
 - The pre-treatment time remains unchanged; The treated time is rescaled by a factor $\exp(\psi)$
- e^{ψ} : acceleration factor

Rank Preserving Structure Failure Time Model (RPSFTM)

Re-censoring

- **Observed World**

- The patient does **not** experience the event; We observe a censoring time C_i ; Beyond C_i , nothing is observed

- **Counterfactual (No-Treatment) World**

- Survival time is transformed to $U_i(\psi)$; The censoring time must also be transformed to $C_i(\psi)$

- **Why Re-censoring Is Required**

- After time transformation, the *observed* censoring time becomes $C_i(\psi)$, which can depend on treatment history → **informative censoring** → **bias**
- Re-censoring restores the independence

$$D_i^*(\psi) = \min \left(\underbrace{C_i}_{\text{original admin censoring}}, \underbrace{C_i \exp(\psi)}_{\text{censoring mapped to no-treatment scale}} \right)$$

Rank Preserving Structure Failure Time Model (RPSFTM)

How Is ψ Estimated

- **Counterfactual survival time:**

$$U_i(\psi) = T_{i,off} + T_{i,on} \boxed{\exp(\psi)}$$

↓
Acceleration factor

- **Goal:** Find ψ that makes the two randomized groups look the same after time transformation.

Rank Preserving Structure Failure Time Model (RPSFTM)

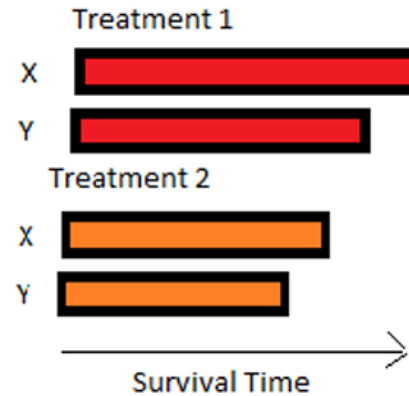
How Is ψ Estimated – G-estimation

- **Step 1: Set a search range for ψ**
- **Step 2: For each candidate ψ , do three things**
 - (A) Compute each subject's counterfactual survival time $U_i(\psi)$
 - (B) Apply re-censoring if necessary
 - (C) Use a survival difference test to test the following hypothesis: $H_0: U_i(\psi) \perp R_i$
- **Step 3: Solve for ψ**
 - Find the value of ψ where the survival should be the same in the two randomized groups.
 - This $\hat{\psi}$ is the g-estimate of the treatment effect (CIs are commonly obtained via bootstrap)

Rank Preserving Structure Failure Time Model (RPSFTM)

Assumptions

- Randomization ensures the treatment and control arms are comparable at baseline.
- In a hypothetical world, the two randomized groups would have the same survival distribution.
- Therefore, treatment assignment affects survival **only through actual treatment exposure**.



- If patient X would live longer than patient Y without treatment, then patient X would also live longer than patient Y if both were treated.
- Treatment rescales time but does not reverse who survives longer.

- Every unit of time on treatment has **the same effect for all patients**
- The treatment effect does **not depend on**:
 - when treatment starts,
 - disease stage at treatment initiation,
 - whether treatment is received at baseline or after switching
- All treated time is rescaled by **one common factor**, $\exp(\psi)$

Rank Preserving Structure Failure Time Model (RPSFTM)

R Code - `rpsftm` package

- https://cran.r-project.org/web/packages/rpsftm/vignettes/rpsftm_vignette.html

- Syntax

```
rpsftm(formula, data, censor_time, subset, na.action, test=survdiff, low_psi=-1, hi_psi=1, alpha=0.05, treat_modifier=1, autoswitch=TRUE, n_eval_z=100, ...)
```

- Dataset (immdef)

A simulated dataset from White et al. (2002), based on a randomized trial comparing immediate versus deferred zidovudine treatment in asymptomatic HIV-infected participants.

	id	def	imm	censyrs	xo	xoyrs	prog	progyrs	entry
1	1	0	1	3	0	0.000000	0	3.000000	0
2	2	1	0	3	1	2.652797	0	3.000000	0
3	3	0	1	3	0	0.000000	1	1.737838	0
4	4	0	1	3	0	0.000000	1	2.166291	0
5	5	1	0	3	1	2.122100	1	2.884646	0
6	6	1	0	3	1	0.557392	0	3.000000	0

- **def**: indicator for assignment to deferred treatment
- **imm**: indicator for assignment to immediate treatment
- **censyrs**: censoring time
- **xo**: indicator of treatment switching
- **xoyrs**: time from randomization to treatment crossover
- **prog**: indicator of disease progression

Rank Preserving Structure Failure Time Model (RPSFTM)

R Code - ITT analysis

```
> fit_cox = coxph(Surv(progyrs, prog) ~ imm, data=immdef)
> summary(fit_cox)
call:
coxph(formula = Surv(progyrs, prog) ~ imm, data = immdef)

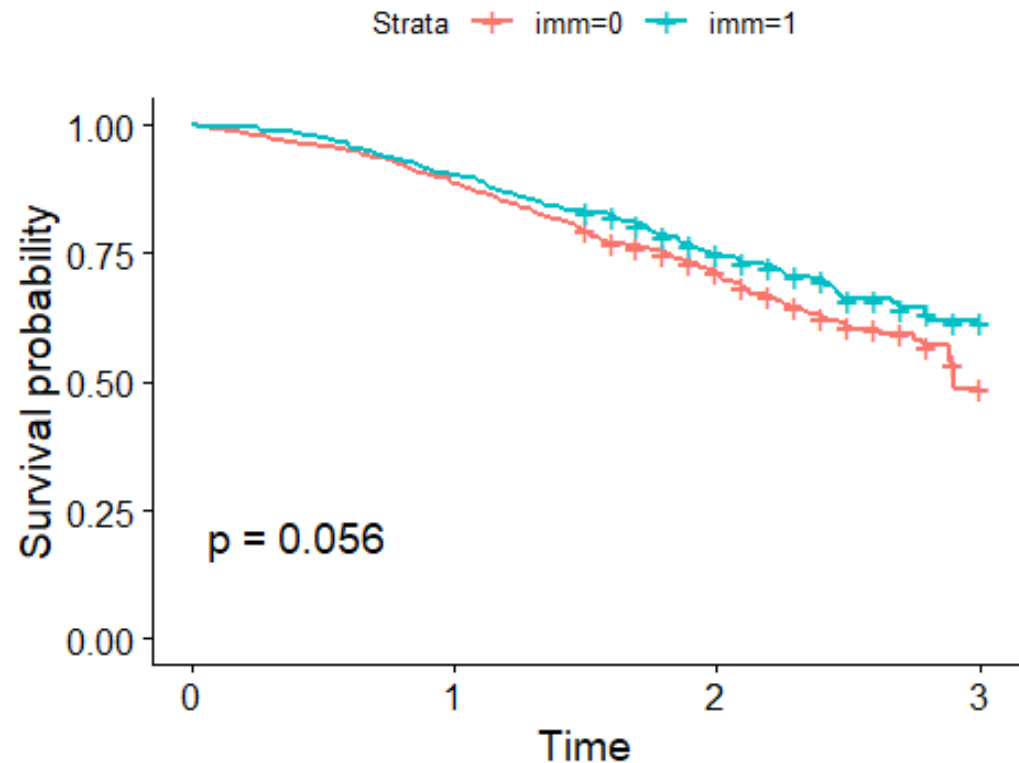
n= 1000, number of events= 312

      coef exp(coef) se(coef)      z Pr(>|z|)
imm -0.2171    0.8048   0.1137 -1.91  0.0561 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

      exp(coef) exp(-coef) lower .95 upper .95
imm    0.8048    1.243    0.6441    1.006

Concordance= 0.525  (se = 0.015 )
Likelihood ratio test= 3.66  on 1 df,   p=0.06
Wald test               = 3.65  on 1 df,   p=0.06
Score (logrank) test = 3.66  on 1 df,   p=0.06
```

```
KM_noadj <- survfit(Surv(progyrs, prog) ~ imm,
data=immdef)ggsurvplot(KM_noadj,data = immdef,pval = T)
```



Rank Preserving Structure Failure Time Model (RPSFTM)

R Code - RPSFTM

```
rx <- with(immdef, 1 - xoyrs/progyrs)
```

```
rpsftm_fit_lr <- rpsftm(formula = Surv(progyrs, prog) ~ rand(imm, rx), data = immdef, censor_time = censyrs)
```

```
  arm  rx.Min. rx.1st Qu. rx.Median  rx.Mean rx.3rd Qu.  rx.Max.
1    0 0.000000 0.000000 0.000000 0.1574062 0.2547779 0.9770941
2    1 1.000000 1.000000 1.000000 1.0000000 1.0000000 1.0000000
call:
rpsftm(formula = Surv(progyrs, prog) ~ rand(imm, rx), data = immdef,
        censor_time = censyrs)
```

	N	Observed	Expected	(O-E) ² /E	(O-E) ² /V
.arm=0	500	142	142	0.000446	0.00089
.arm=1	500	143	143	0.000444	0.00089

Chisq= 0 on 1 degrees of freedom, p= 1

```
psi: -0.1813226
exp(psi): 0.8341662
```

```
> rpsftm_fit_lr$psi
[1] -0.1813226
> rpsftm_fit_lr$CI
[1] -0.34984001 0.002287789
```

Rank Preserving Structure Failure Time Model (RPSFTM)

SAS Code

- Key steps

```
/* Treatment assignment logic */
if imm = 1 then do;
    t_on = progyrs;
    t_off = 0;
end;

if imm = 0 then do;
    if xo = 1 then do;
        t_on =
        t_off =
    end;
    else do;
        t_on =
        t_off =
    end;
end;

/* Total observed time */
T = t_off + t_on;

/* Counterfactual no-treatment time */
U = t_off + exp(&psi) * t_on;

/* Re-censoring */
cut_U = min(exp(&psi) * censyrs, censyrs);

if prog = 1 then do;
    if U > cut_U then do;

psi: -0.181
exp(psi): 0.834
Confidence Interval, psi -0.349 0.002
Confidence Interval, exp(psi) 0.705 1.002

    end;
end;

/* Grid search over psi */
data step1;
    set immdef_dataset;
    do psi_search = -0.5 to 1.0 by 0.001;
        %find_U(psi = psi_search);
        output;
    end;
run;

data=step1;
arch;
event_U(0);
mcat / test=logrank;
```

over ψ

Identify $\hat{\psi}$ by log-rank test

Split each subject's follow-up time into time on treatment and time off treatment

Construct Counterfactual survival time and re-censoring

Rank Preserving Structure Failure Time Model (RPSFTM)

Pros and Cons

Pros

- Does not rely on covariate information, thereby relaxing the *no unmeasured confounding* assumption.
- Remains applicable even with very high levels of treatment switching, including settings where all patients in an arm switch (a scenario where MSM and IPCW may fail).

Cons

- This method is only suitable for one-way switching.
- Increased SE by re-censoring and re-sampling (bootstrap and permutation) on the whole RPSFTM process in computing 95% CI and p-value
- Computation resources intensive due to re-sampling on the whole process
- G-estimation grid search may end up with multiple roots for $\hat{\psi}$
- “**Common treatment effect**” assumption is a strong assumption, which may be not true for some oncology trials.

Two Stage Estimation

Two Stage Estimation

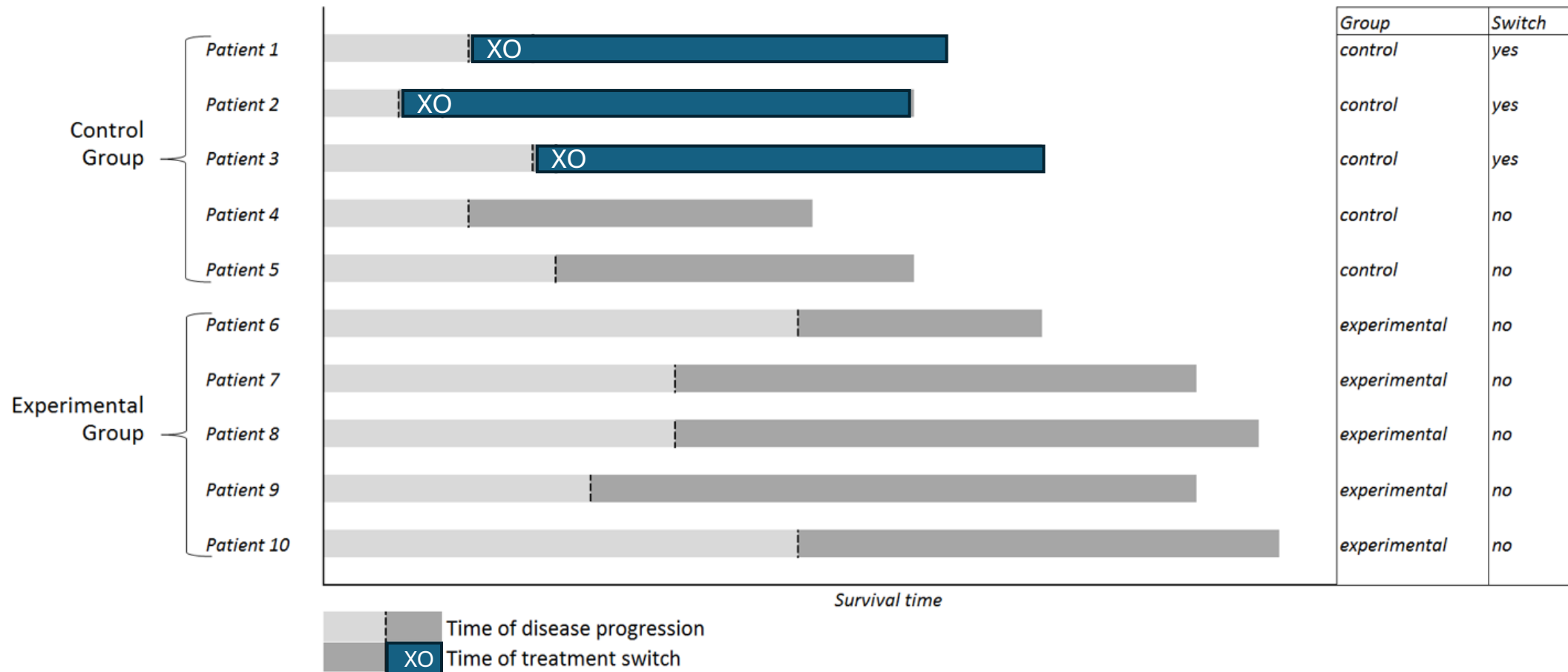
It is a re-scaling approach, like the RPSFTM method.
But do not use “common treatment” assumption.

TSE relies on the following 3 assumptions:

- No unmeasured confounding (at 1st and 2nd baseline)
- OS data follow a parametric distribution
- No gap time b/t progression and treatment switch

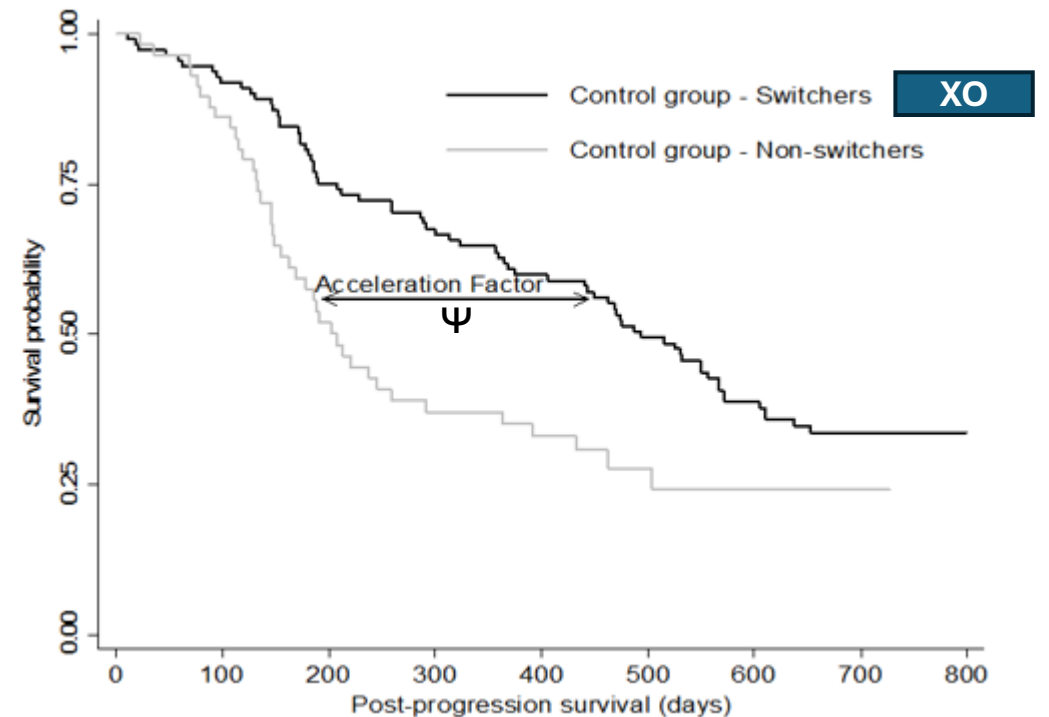
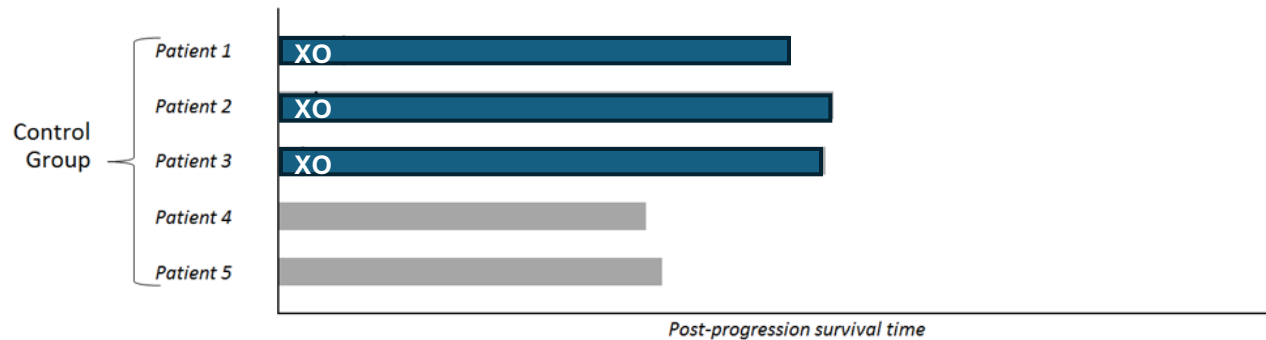
Step 1:

- Identify secondary baseline for each patient



Step 2:

- Compare post-secondary baseline survival in control group non-switchers with control group switchers using an AFT model



AFT model

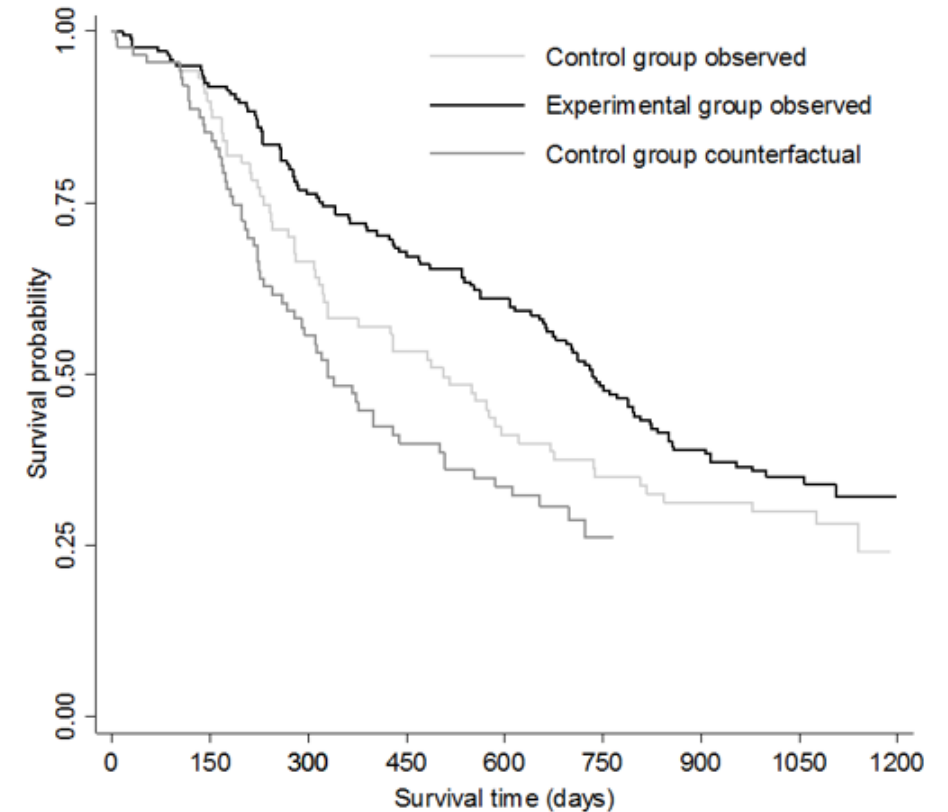
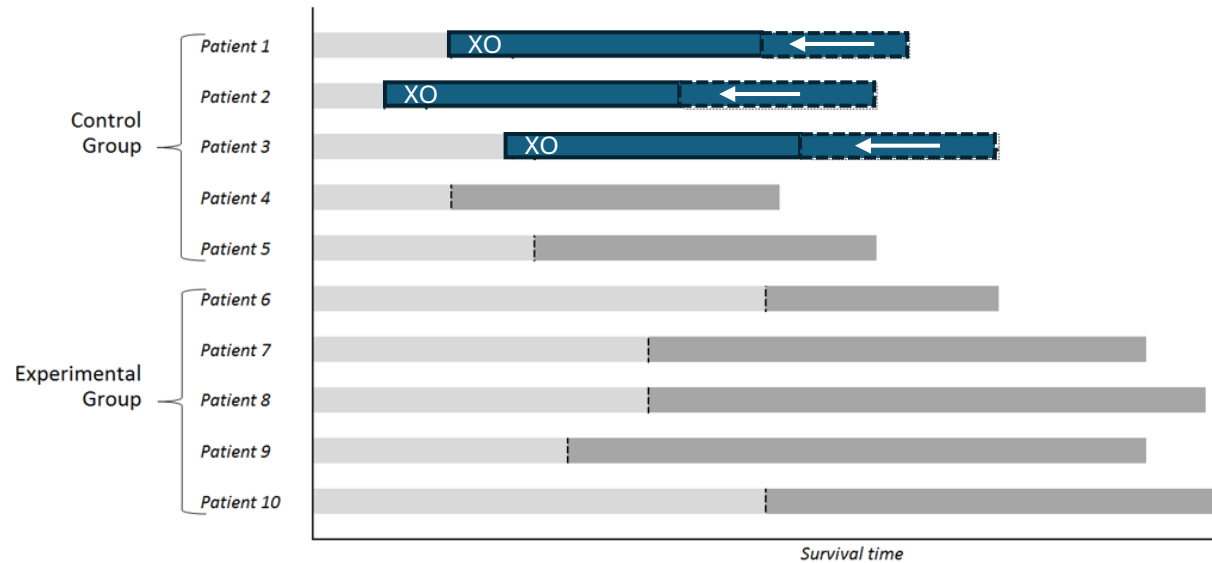
In survival analysis, instead of focusing on the *hazard function* (like a Cox proportional hazards model), an **AFT model looks at the *time scale*** and assumes covariates multiply the survival time.

The model assumes the effect of covariates is to **accelerate or decelerate the life course of the event of interest** (i.e., make the event happen sooner or later).

Unlike Cox models, AFT models require you to specify a **baseline distribution** for survival times

Step 3:

- Shrink survival times in switchers according to acceleration factor, and compare to experimental group to obtain hazard ratio adjusted for switching



Accommodating informative censoring

The TSE method involves adjusting censoring times in switchers, but not in non-switchers. This will result in informative censoring if there is an association between switching and prognosis.

Use IPCW Method to accommodate informative censoring.

Limitations

Assumption 1:

No unmeasured confounding (at 1st and 2nd baseline)

- It cannot be precisely true, but measures can be taken to make it 'approximately true' at the trial design stage.
- In the study protocol, devote a section for 1st and 2nd baseline data collection

Limitations

Assumption 2:

OS data follow a parametric distribution

- It is not easy to find a parametric distribution which fits observed data perfectly

Limitations

Assumption 3:

No gap time b/t progression and treatment switch

- Gap time often exists in most relapsed patients
- Negative gap time

Other extreme cases:

- Almost all relapsed patients experienced treatment switching, which leads to unreliable Ψ estimate
- Some patients experienced treatment switching without disease progression, estimated Ψ may not be applicable to this set of patients

R code

library(trtswitch)

```
> head(shilong,10)
  id tstart tstop event  agerand sex.f tt_Lnum rmh_alea.c pathway.f bras.f ps ttc tran dpd dco ady dcut pd co
1  1      0   28     0  76.62628  Male      3          0 PI3K/AKT/mTOR  CT  1  0   0  28  31 145 1228 1 TRUE
2  1     28  133     0  76.62628  Male      3          0 PI3K/AKT/mTOR  CT  1  1   0  28  31 145 1228 1 TRUE
3  1    133  145     1  76.62628  Male      3          0 PI3K/AKT/mTOR  CT  2  1   0  28  31 145 1228 1 TRUE
4  2      0   34     0  65.10541 Female     5          0 PI3K/AKT/mTOR  MTA  1  0   0  34  NA  64 1114 1 FALSE
5  2     34   64     1  65.10541 Female     5          0 PI3K/AKT/mTOR  MTA  3  1   0  34  NA  64 1114 1 FALSE
6  3      0   20     0  36.82341 Female     2          0          HR  CT  1  1   0 106 127 287 1227 1 TRUE
7  3     20   48     0  36.82341 Female     2          0          HR  CT  1  0   0 106 127 287 1227 1 TRUE
8  3     48   70     0  36.82341 Female     2          0          HR  CT  0  1   0 106 127 287 1227 1 TRUE
9  3     70  127     0  36.82341 Female     2          0          HR  CT  1  1   0 106 127 287 1227 1 TRUE
10 3    127  155     0  36.82341 Female     2          0          HR  CT  2  1   1 106 127 287 1227 1 TRUE
```

Bras.f: randomized treatment arm

Time dependent covariates

ps The ECOGperformance status

ttc The presence of concomitant treatments

tran The use of platelet transfusions

dpd: relative day of progression

dco : relative day of treatment switching

ady: relative day of last news/follow-up

pd: indicator for progression

co: indicator for treatment switching

```

# the eventual survival time
shilong1 <- shilong %>%
  arrange(bras.f, id, tstop) %>%
  group_by(bras.f, id) %>%
  slice(n()) %>%
  select(-c("ps", "ttc", "tran"))

#bras.f The patient's randomized arm, either MTA or CT
#ps The ECOGperformance status
#ttc The presence of concomitant treatments
#tran The use of platelet transfusions

# the last value of time-dependent covariates before pd
shilong2<- shilong %>%
  filter(pd == 0 | tstart <= dpd) %>%
  arrange(bras.f, id, tstop) %>%
  group_by(bras.f, id) %>%
  slice(n()) %>%
  select(bras.f, id, ps, ttc, tran)

# combine baseline and time-dependent covariates
shilong3 <- shilong1 %>%
  left_join(shilong2, by = c("bras.f", "id"))

```

```

> subset(shilong, shilong$id==1)
  id tstart tstop event  agerand sex.f tt_Lnum rmh_alea.c pathway.f bras.f ps ttc tran dpd dco ady dcut pd  co
1  1      0    28     0 76.62628 Male      3          0 PI3K/AKT/mTOR   CT  1  0  0  28  31 145 1228  1 TRUE
2  1     28   133     0 76.62628 Male      3          0 PI3K/AKT/mTOR   CT  1  1  0  28  31 145 1228  1 TRUE
3  1    133   145     1 76.62628 Male      3          0 PI3K/AKT/mTOR   CT  2  1  0  28  31 145 1228  1 TRUE
> subset(shilong3, shilong3$id==1)
# A tibble: 1 x 19
# Groups:   bras.f, id [1]
   id tstart tstop event agerand sex.f tt_Lnum rmh_alea.c pathway.f bras.f dpd dco ady dcut pd co ps ttc tran
  <int> <dbl> <dbl> <int> <dbl> <fct> <int> <dbl> <fct> <fct> <dbl> <dbl> <dbl> <dbl> <dbl> <lg1> <dbl> <dbl> <dbl>
1    1    133   145     1  76.6 Male      3          0 PI3K/AKT/mTOR CT      28  31  145  1228  1 TRUE    1    1    0

```

```
fit1 <- tsesimp(  
  data = shilong3, id = "id", time = "tstop", event = "event",  
  treat = "bras.f", censor_time = "dcut",  
  pd = "pd", pd_time = "dpd",  
  swtrt = "co", swtrt_time = "dco",  
  base_cov = c("agerand", "sex.f", "tt_Lnum", "rmh_alea.c", "pathway.f"),  
  base2_cov = c("agerand", "sex.f", "tt_Lnum", "rmh_alea.c", "pathway.f", "ps", "ttc", "tran"),  
  aft_dist = "weibull", alpha = 0.05, recensor = TRUE, swtrt_control_only = FALSE, offset = 1, boot = FALSE)
```

```

> # control group
> fit1$fit_aft[[1]]$fit$parest[, c("param", "beta", "sebeta", "z")]
      param      beta      sebeta      z
1 (Intercept)  5.572896483  0.758612068  7.3461743
2      swtrt    1.067653070  0.237997891  4.4859770
3      agerand  0.001927251  0.007493015  0.2572064
4 sex.fFemale  0.316148630  0.240086216  1.3168129
5      tt_Lnum -0.023248007  0.040491209 -0.5741495
6      rmh_alea.c -0.894680233  0.193398728 -4.6260916
7      pathway.fHR -0.484778956  0.358671753 -1.3515950
8 pathway.fPI3K.AKT.mTOR -0.604986176  0.382244440 -1.5827207
9          ps    -0.548628111  0.136464658 -4.0202945
10         ttc    0.050010291  0.269239313  0.1857466
11         tran  0.437778730  0.385005847  1.1370703
12      Log(scale) -0.428099622  0.105960290 -4.0401892
> fit1$psi
[1] -1.067653
> # experimental group
> fit1$fit_aft[[2]]$fit$parest[, c("param", "beta", "sebeta", "z")]
      param      beta      sebeta      z
1 (Intercept)  4.819297083  0.760270308  6.3389258
2      swtrt    0.983747428  0.271723182  3.6204030
3      agerand  0.001253863  0.009513484  0.1317985
4 sex.fFemale  0.395790503  0.237599761  1.6657866
5      tt_Lnum  0.066813220  0.051717213  1.2918952
6      rmh_alea.c -0.714216404  0.209567497 -3.4080495
7      pathway.fHR -0.301948752  0.296989948 -1.0166969
8 pathway.fPI3K.AKT.mTOR -0.076210121  0.281075448 -0.2711376
9          ps    -0.401333183  0.114040814 -3.5192066
10         ttc    0.414584864  0.377881586  1.0971290
11         tran -0.615854802  0.390436760 -1.5773484
12      Log(scale) -0.414571271  0.116718016 -3.5519047
> fit1$psi_trt
[1] -0.9837474

```

```

> subset(shilong3, shilong3$id %in% c(1,10))
# A tibble: 2 x 19
# Groups:   bras.f, id [2]
   id tstart tstop event agerand sex.f tt_Lnum rmh_alea.c pathway.f bras.f dpd dco ady dcut pd co ps ttc tran
  <int> <dbl> <dbl> <int> <dbl> <fct> <int> <dbl> <fct> <fct> <dbl> <dbl> <dbl> <dbl> <dbl> <lgl> <dbl> <dbl> <dbl>
1    1   133   145     1   76.6 Male      3      0 PI3K/AKT/mTOR CT      28    31   145   1228     1 TRUE     1     1     0
2   10     7    37     1   47.2 Female    8      1 PI3K/AKT/mTOR CT      15    NA    37   1211     1 FALSE     2     1     0
> subset(fit1$data_outcome, fit1$data_outcome$id %in% c(1,10))
   id t_star d_star treated agerand tt_Lnum rmh_alea.c bras.f sex.f pathway.f
101 1 67.57011     1      0 76.62628      3      0      2 Male PI3K/AKT/mTOR
105 10 37.00000     1      0 47.21492      8      1      2 Female PI3K/AKT/mTOR
> |

```

```
> #fit the outcome Cox model and compare the treatment hazard ratio estimate with the reported.
> fit1$fit_outcome$parest[, c("param", "beta", "sebeta", "z")]
      param      beta      sebeta      z
1      treated -0.0914777971 0.184469500 -0.49589660
2      agerand  0.0045435260 0.008026488  0.56606649
3      sex.fFemale -0.4181004068 0.191327703 -2.18525806
4      tt_Lnum -0.0008255737 0.036590479 -0.02256253
5      rmh_alea.c  0.6881577040 0.191026441  3.60242121
6      pathway.fHR  0.1580245496 0.297687989  0.53083952
7 pathway.fPI3K.AKT.mTOR 0.1873258331 0.303906799  0.61639237
>
>
> c(fit1$hr, fit1$hr_CI)
[1] 0.9125816 0.6356982 1.3100637
```

SAS

Data process:

From longitudinal long data format to one record per row for AFT modeling.

Derive phi

```
ods select ParameterEstimates;
proc lifereg data=pd2 outest=phi2;
  class xover indctreg asctresp lyticbone_model mrd_model time_induct_trt
    cytocat CRECEL_finalmodel;
  model osday*die(0)=xover age_model region_model
    CYTOCAT ecog_model baseCORRCA_model duraEx_model pd locfspep locfplat
    ASCTRESP baseHGB_model
    baseLDH_model plascell_model locfupep /*locfhgb race_model B2MGID basePLAT_model meltype_model*/
  /dist=weibull;
  * with smallest AIC;
  title "Two-stage model for OS, xover=2 is for non-xover";
  title2 "Stage 1: Use proc lifereg, get slowdown parameter estimate for Xover";
  title3 "Use covarite to adjust, assume weibull survival time distribution";
  title4 "Drop those covariates for which p-value is too close to 1";
run;

data _null_;
  set phi2;
  call symput('phiweb1', put(xover1,best.));
  put xover1;
run;
```

Use AIC/BIC to pick which parametric distribution fits better

The LIFEREG Procedure

Analysis of Maximum Likelihood Parameter Estimates								
Parameter		DF	Estimate	Standard Error	95% Confidence Limits		Chi-Square	Pr > ChiSq
Intercept		1	4.5090	6.2352	-7.7118	16.7298	0.52	0.4696
xover	1	1	0.9093	0.2257	0.4668	1.3517	16.23	<.0001
xover	2	0	0.0000	.	.	.	.	.
age_model		1	-0.2477	0.1930	-0.6260	0.1305	1.65	0.1993
region_model		1	-0.3454	0.3190	-0.9706	0.2798	1.17	0.2789
CYTOCAT	High Risk	1	-0.5123	0.2753	-1.0519	0.0273	3.46	0.0628
CYTOCAT	Standard Risk	1	-0.1257	0.2789	-0.6724	0.4209	0.20	0.6522
CYTOCAT	Unclassifiable	0	0.0000	.	.	.	.	.
ecog_model		1	-0.5772	0.4964	-1.5501	0.3957	1.35	0.2449
baseCORRCA_model		1	1.7394	2.5697	-3.2971	6.7759	0.46	0.4985
duraEx_model		1	0.0582	0.0152	0.0283	0.0881	14.58	0.0001
PD		0	0.0000	.	.	.	.	.
LOCFSEPEP		1	0.0663	0.0231	0.0210	0.1115	8.24	0.0041
LOCFPLAT		1	0.0006	0.0015	-0.0024	0.0036	0.17	0.6791
ASCTRESP	CR or VGPR	1	-0.1940	0.2715	-0.7263	0.3382	0.51	0.4749
ASCTRESP	PR	0	0.0000	.	.	.	.	.
baseHGB_model		1	-0.0085	0.0091	-0.0263	0.0093	0.88	0.3488
baseLDH_model		1	0.0016	0.0024	-0.0031	0.0063	0.45	0.5002
plascell_model		1	-0.2180	0.1971	-0.6043	0.1683	1.22	0.2687
LOCFUPEP		1	-0.0030	0.0026	-0.0082	0.0021	1.33	0.2480
Scale		1	0.6137	0.0735	0.4853	0.7761		
Weibull Shape		1	1.6295	0.1952	1.2885	2.0608		

For switchers, after treatment switch, recompute survival time as $U = T_c + T_a / \exp(\phi)$, ϕ is from AFT model

```
data control;  
  set control9;  
  by usubjid analdy;  
  
  if xover=0 then osday2=analdy;  
  else if xover=1 then osday2=xoverdy + round((analdy-xoverdy)/exp(&phiweb1.));  
  analdy=osday2;  
run;
```

```
proc phreg data=main_w2b covs(aggregate);  
  class trt01p(ref='Placebo') indctreg asctresp stgenrl;  
  model (start, stop)*die(0)=trt01p /rl;  
  strata indctreg asctresp stgenrl;  
  id usubjid;  
  weight stabw;  
  
  title "TSE+IPCW Cox model for OS";  
  title2 "Stage 2: Using proc phreg, counting process data format for multiple records";  
  title3 "IPCW is used to accomodate informative censoring in the control arm";  
run;
```

The PHREG Procedure

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
TRT01P	1	8.8052	0.0030

Analysis of Maximum Likelihood Estimates											
Parameter		DF	Parameter Estimate	Standard Error	StdErr Ratio	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
TRT01P	Active	1	-0.58190	0.19610	0.965	8.8052	0.0030	0.559	0.381	0.821	Planned Treatment for Period 01 Active

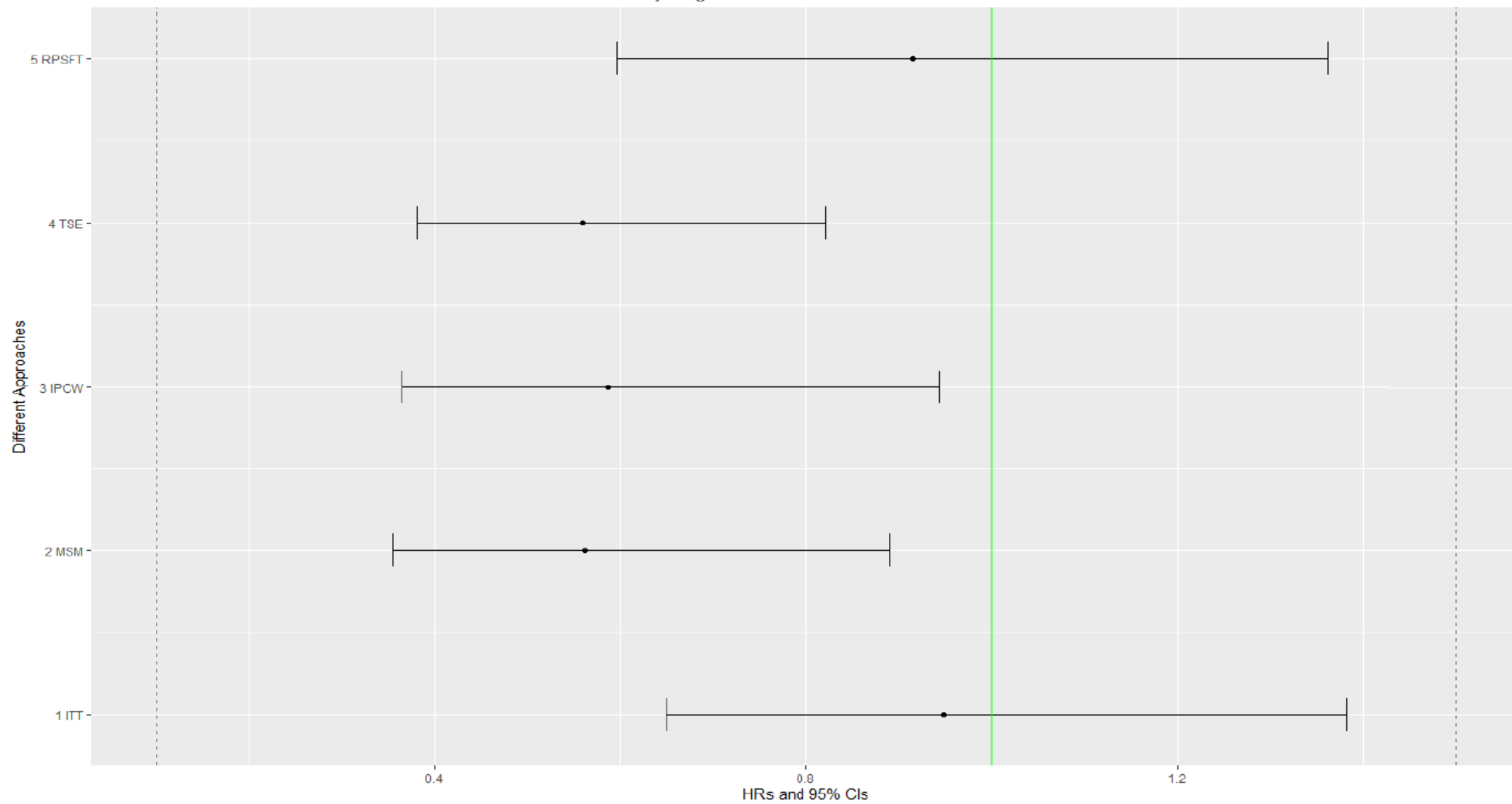
Summary

Treatment switching complicates survival analyses because standard ITT can underestimate effects.

- MSM/IPCW re-weights data to adjust for the informative censoring due to switching.
- RPSFTM re-scales survival times assuming a constant effect across switchers and non-switchers.
- TSE uses an AFT model in two stages to construct adjusted survival times.

Method	Key assumption	Strength	Limitations
RPSFTM	Common treatment effect ▪ May not be true for oncology RCT	➤ Needs no covariate information ➤ No limit on switch proportions	➤ “common treatment effect” may be a too strong assumption ➤ Increased SE by re-censoring and resampling ➤ Shortened placebo arm in adjusted KM plot ➤ Computation resources intensive ➤ Grid search may result in multiple roots for ψ_{hat}
TSE	➤ No unmeasured confounding ▪ manageable for RCT via pre-specification in SAP ➤ OS data follow a parametric distribution ➤ No gap time b/t progression and crossover	➤ No common treatment effect assumption ➤ SE may be estimated via data driven robust variance ➤ Data may be summarized via adjusted Stat. and KM curve like ITT	➤ key assumptions may be violated ➤ Patients' switching time may be prior to relapse ➤ Patients may switch without relapse ➤ RCT data may possess extra multiple deviations from ideal TSE application ➤ Need to develop Modified TSE approach
IPCW	no unmeasured confounding ▪ manageable for RCT via pre-specification in SAP	➤ No common treatment effect assumption ➤ SE may be estimated via data driven robust variance ➤ Data may be summarized via adjusted Stat. and KM curve like ITT ➤ May be combined with other methods to deal with informative censoring ➤ Model setting is simpler compared to MSM	➤ Wider 95% Cis due to information loss from censoring the post-treatment data ➤ Need to meet positivity and consistency assumption ➤ May fail if too many (or too few) patients switched ➤ High demand on non-missing covariate data overtime
MSM	no unmeasured confounding ▪ manageable for RCT via pre-specification in SAP	➤ No common treatment effect assumption ➤ SE may be estimated via data driven robust variance ➤ Uses more data than IPCW	➤ Need to meet positivity and consistency assumption ➤ May fail if too many (or too few) patients switched ➤ High demand on non-missing covariate data

HR Estimates and 95% CIs from Different Methods Before and After Adjusting for Crossover



Best practice recommendation

- Up front planning
- Pre-specify analysis details in the SAP, including a complete list of confounding covariates and variable selection method, etc.
- Early engagement with regulatory agencies
- Number and percentage of patients with treatment crossover or subsequent therapy need to be summarized by treatment arm
- More than one method (better if belonging to different classes) needs to be used to demonstrate the robustness of the causal OS treatment benefit, e.g., TSE and MSM